Online Handwritten Chinese Character Recognition via Flow Field Features and Convolutional Networks

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Abstract

While extensive research has been done on classifying native speaker Mandarin handwriting, there is far less research on classifying second-language learner handwriting. This project addresses this gap by presenting a system for classifying online handwritten Chinese characters trained on the CASIA-OLHWDB dataset across 500 character classes. We evaluate two modeling approaches: a logistic regression baseline operating on 512-dimensional flow field feature vectors derived from Liu et al. (2012), and a convolutional neural network (CNN) that treats the same flow field vectors as spatial direction maps. In addition to the standard CASIA dataset, both models are evaluated on an augmented dataset that includes omitted strokes, mimicking a new learner's frequent mental lapses during writing. The logistic regression baseline achieves 86.3% and 79% test accuracy on the original and augmented datasets respectively, demonstrating the strong discriminative power of hand-engineered features. The CNN exploits the spatial structure of the direction maps and achieves 92.5% and 87.8% test accuracy, showing that local convolutional operations over direction channels provide meaningful gains over a linear classifier on the same features.

1 Introduction

Online handwritten Chinese character recognition (OLHCR) is a challenging machine learning problem due to the large number of character classes, high variability within class due to writing style. However, because the temporal nature of online stroke data, which, unlike offline handwriting, has access to the full temporal trajectory of the pen — including stroke order, direction, and speed — it is a rich source of discriminative information.

Chinese is particularly challenging: standard usage requires knowledge of thousands of characters, each composed of a variable number of strokes that writers may execute in different orders and with varying levels of detail. Recognizing even a subset of 500 characters already constitutes a demanding 500-class classification problem. Beyond native speakers, second-language learners present an additional challenge: they frequently omit, misorder, or reverse strokes, producing character instances that deviate structurally from the training distribution.

In this work, we train and evaluate two approaches on the CASIA-OLHWDB benchmark (1). First, we implement the flow field feature extraction pipeline of Liu et al. (2), which encodes directional stroke information into a 512-dimensional feature vector organized as an $8 \times 8 \times 8$ direction map, and train a logistic regression classifier on top. Second, we design a CNN that treats this direction map as a spatial feature volume and applies convolutional filters to capture local interactions between direction channels and spatial grid cells — interactions that logistic regression cannot model. We evaluate both models on held-out test data and study the impact of stroke-level data augmentation on model robustness to learner-style writing.

2 Background and Related Work

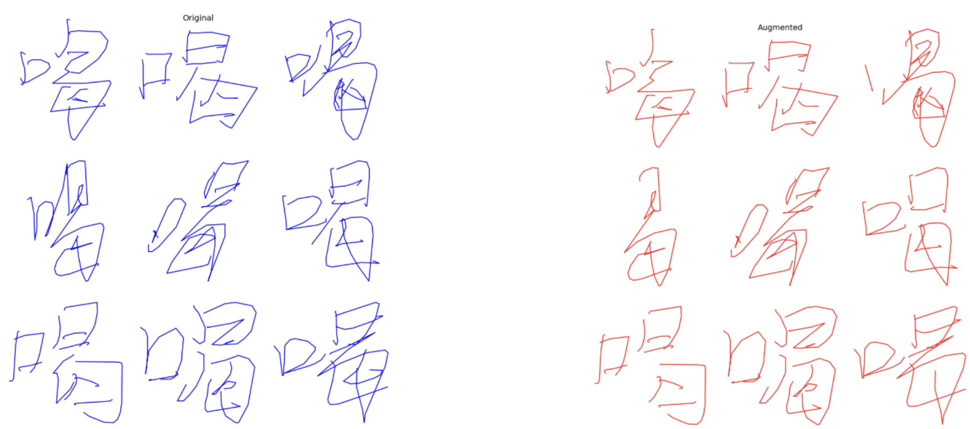
2.1 Online Handwriting Recognition

Online handwriting recognition differs from offline recognition in that it processes sequences of (x, y) coordinates sampled at discrete time steps, grouped into strokes separated by pen lifts. This sequential structure has historically been exploited through dynamic time warping (DTW), hidden Markov models (HMMs), and recurrent neural networks (3).

For Chinese specifically, the large class set and stroke complexity motivate feature engineering approaches that compress stroke trajectories into compact, discriminative representations. Liu et al. (2) proposed direction map features computed over a spatial grid, which remain competitive with learned features on standard benchmarks.

2.2 Dataset

We use the CASIA-OLHWDB dataset (1), specifically the Pot1.0 and Pot1.1 splits stored in the POT binary format. Each file contains samples for individual writers, with each sample consisting of a sequence of strokes, and each stroke a sequence of (x, y) coordinate pairs. We filter to a fixed vocabulary of the most common 500 Chinese characters and merge data across both dataset versions to maximize training coverage.



(a) Original samples (b) Augmented samples

Figure 1: Example characters from CASIA-OLHWDB before and after stroke-level augmentation. Augmented samples exhibit stroke reordering, occasional stroke removal, and reversed stroke directions, simulating the writing patterns of second-language learners.

After merging, we perform a 90/10 train/validation split per character, yielding 689,785 training samples and 76,938 validation samples across 500 classes. The held-out test set is drawn from the separate Pot1.0Test and Pot1.1Test splits.

3 Feature Extraction

3.1 Resampling and Normalization

Before computing features, each stroke is resampled to uniform arc-length spacing with interval $\delta = 2.0$, ensuring that feature computation is invariant to variations in writing speed. Coordinates are then normalized using Pseudo-2D Bi-Moment Normalization (P2DBMN) (2), which aligns characters by their spatial distribution and removes absolute position and scale.

3.2 Direction Maps

Following Liu et al. (2), we compute direction map features by projecting stroke tangent directions onto 8 chaincode directions using the parallelogram rule. The character bounding box is divided into an 8×8 spatial grid, and direction votes are accumulated per cell. Pen lifts between strokes are assigned half-weight to adjacent cells. This yields an $8 \times 8 \times 8 = 512$-dimensional direction map, which we interpret as 8 direction channels over an 8×8 spatial grid.

Each direction channel is smoothed with a Gaussian filter ($\sigma = 1.0$). Finally, we apply a Box-Cox transform ($\lambda = 0.5$, i.e., square root) to compress the dynamic range of the resulting feature vector.

3.3 Summary

The full feature extraction pipeline maps a variable-length stroke sequence to a fixed 512-dimensional vector, equivalently a $C \times H \times W = 8 \times 8 \times 8$ feature volume:

$$f : \{(x_t, y_t)\}_{t=1}^{T} \;\longrightarrow\; \mathbf{v} \in \mathbb{R}^{512} \;\equiv\; \mathbf{V} \in \mathbb{R}^{8\times8\times8} \tag{1}$$

4 Data Augmentation

To improve model robustness to writing style variation — and specifically to simulate second-language learner writing — we apply stroke-level augmentation during training. Three perturbation types are applied stochastically to each sample:

Stroke removal: Each stroke is independently removed with probability $p_{\text{remove}} = 0.05$. Removal cascades to adjacent strokes with the same probability, simulating the tendency of learners to omit structurally related strokes together.

Stroke swapping: Adjacent stroke pairs are swapped with probability $p_{\text{swap}} = 0.1$, simulating variation in stroke order common among learners still developing motor memory.

Stroke reversal: Individual strokes are reversed in direction with probability $p_{\text{reverse}} = 0.05$, simulating incorrect stroke direction, a frequent error among non-native writers.

We generate $k = 1$ augmented copies per original sample, doubling the effective dataset size while preserving all original samples. Feature extraction is then applied to both original and augmented stroke sequences, producing a matched pair of feature datasets for controlled evaluation.

5 Models

5.1 Logistic Regression Baseline

As a strong baseline, we train logistic regression directly on the 512-dimensional flow field feature vectors. Despite its simplicity, logistic regression is a natural choice here: the Liu et al. features are explicitly designed to be linearly separable, and the model provides a clean measure of what hand-engineered features can achieve without nonlinear interactions.

5.1.1 Model

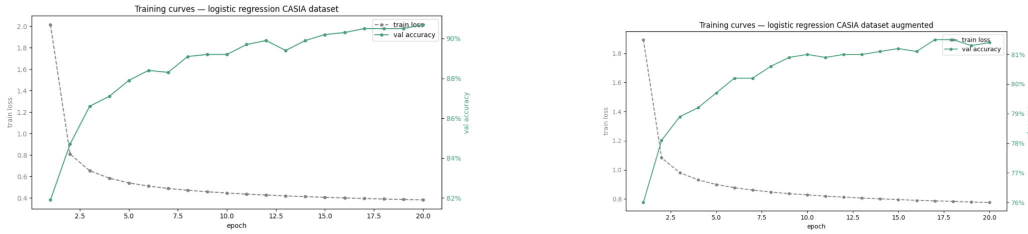
The model computes the posterior probability of class c given feature vector $\mathbf{v}$ as:

$$P(y = c \mid \mathbf{v}) = \frac{\exp(\mathbf{w}_c^{\top}\mathbf{v} + b_c)}{\sum_{c'} \exp(\mathbf{w}_{c'}^{\top}\mathbf{v} + b_{c'})} \tag{2}$$

with parameters $\mathbf{W} \in \mathbb{R}^{500\times512}$ and $\mathbf{b} \in \mathbb{R}^{500}$, totaling 256,500 parameters. We minimize cross-entropy loss using Adam with learning rate 10^{-3} and batch size 256.

5.1.2 Results

The logistic regression baseline achieves **86.3%** test accuracy on the original dataset and **79%** on the augmented dataset, demonstrating that the Liu et al. flow field features are highly discriminative for this task. The drop in accuracy on the augmented dataset reflects the challenge of recognizing characters with missing or reordered strokes.



(a) Training loss and validation accuracy

(b) Per-class accuracy distribution

Figure 2: Logistic regression training dynamics and per-class results on the validation set.

5.2 Convolutional Neural Network

Our second model is a CNN that operates on the same 512-dimensional flow field vectors, reshaped into an $8\times8\times8$ direction map. The key motivation is that logistic regression treats the 512 dimensions as independent features, ignoring the spatial structure of the grid. A CNN can detect local patterns — for example, a horizontal stroke in the upper-left region coinciding with a downward stroke in the center — that are invisible to a linear classifier.

5.2.1 Architecture

The input vector $\mathbf{v} \in \mathbb{R}^{512}$ is reshaped to $\mathbf{V} \in \mathbb{R}^{8\times8\times8}$, treating the 8 direction channels as $C = 8$ input channels over an 8×8 spatial grid. Three convolutional blocks with batch normalization and ReLU activations progressively increase the channel depth:

$$\mathbf{Z}^{(1)} = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times3}^{8\to32}(\mathbf{V}))) \tag{3}$$

$$\mathbf{Z}^{(2)} = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times3}^{32\to64}(\mathbf{Z}^{(1)}))) \tag{4}$$

$$\mathbf{Z}^{(3)} = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times3}^{64\to128}(\mathbf{Z}^{(2)}))) \tag{5}$$

All convolutions use 3×3 kernels with padding 1 to preserve spatial dimensions. Adaptive average pooling then reduces the spatial dimensions from 8×8 to 4×4, yielding a $128 \times 4 \times 4 =$ 2,048-dimensional feature vector. A two-layer MLP with dropout ($p = 0.3$) maps this to the 500-class prediction:

$$\hat{y} = \mathbf{W}_2\,\mathrm{ReLU}(\mathbf{W}_1\,\mathrm{Flatten}(\mathbf{Z}^{(3)}) + \mathbf{b}_1) + \mathbf{b}_2 \tag{6}$$

5.2.2 Training

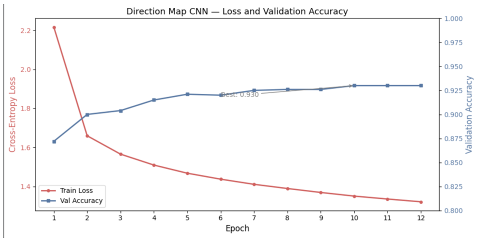
The model is trained for 13 epochs using Adam with learning rate 10^{-3} and a cosine annealing scheduler. Cross-entropy loss with label smoothing ($\epsilon = 0.1$) is used to improve calibration. Mixed-precision training is applied via PyTorch's `GradScaler` on an NVIDIA T4 GPU. Batch size is 256 for training and 512 for validation.

5.2.3 Results

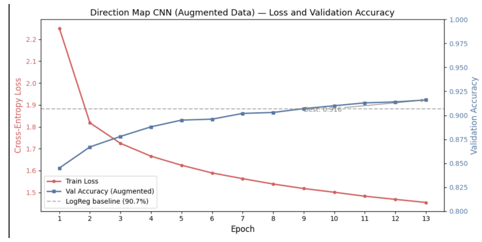
The CNN achieves **92.5%** test accuracy on the original dataset and **87.8%** on the augmented dataset. Training converges rapidly, with most accuracy gains occurring in the first 10 epochs.

6 Results

Both models operate on identical 512-dimensional flow field features, so the difference in performance directly measures the value of modeling spatial interactions between direction channels and grid cells. The CNN's convolutional filters can detect co-occurrences of stroke directions across adjacent spatial regions that are invisible to logistic regression's flat linear classifier.



(a) Training loss and validation accuracy



(b) Per-class accuracy distribution

Figure 3: CNN training dynamics and per-class results on the validation set.

Table 1: Model comparison on the CASIA-OLHWDB test set (500 classes).

Model	Input	Parameters	Val Acc	Test Acc	Test Acc (Aug.)
Logistic Regression	Flow field (512-d)	256,500	90.7%	86.3%	79.0%
CNN	Direction map ($8 \times 8 \times 8$)	1,400,724	93%	92.5%	87.8%

The augmented test set reveals how well each model generalizes to learner-style writing with missing or reordered strokes. The drop in accuracy from the original to the augmented test set measures each model's sensitivity to structural perturbations, which is the key practical question for a second-language writing recognition system. While both models performed comparatively poorly on the augmented dataset, the CNN performed less poorly, indicating that the CNN was learning more thorough structural patterns in the characters, and is more resilient to stroke omissions.

7 Conclusion

We evaluated two approaches to online Chinese character recognition on the CASIA-OLHWDB dataset, with a focus on robustness to the imperfect writing patterns of second-language learners. A logistic regression classifier operating on Liu et al. flow field features achieves 86.3% test accuracy on the original dataset and 79% on augmented data, demonstrating the power of hand-engineered directional representations. A CNN operating on the same features as a spatial $8 \times 8 \times 8$ direction map exploits local structure invisible to logistic regression, achieving 87.8% test accuracy.

Both models are evaluated on an augmented dataset simulating learner writing via stroke removal, reordering, and reversal. The relative robustness of each model on this benchmark has practical implications for deploying Chinese character recognition systems in educational settings for second-language learners.

Future work could incorporate more data from true second-language learners, which would better represent their writing habits than stroke omissions from native writers. This could be added to the native learners' data and the two could be used as a hybrid dataset. This would enable more direct evaluation of the target use-case.

References

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